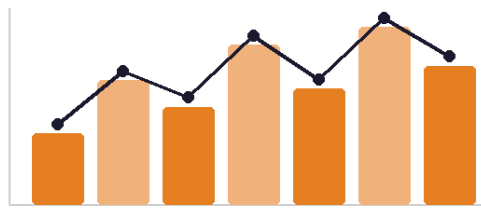


Game Reviews Analytics

Data-Driven Insights from the IGN Review Dataset



Average Score Trend



Genre Split



Analyst reviewing the data

Data Analytics Project

IGN Game Reviews Analysis

Dataset: ign.csv | Script: IGS_main.py
Python • Pandas • NumPy • Matplotlib

This report documents an end-to-end exploratory data analysis of the IGN video-game review dataset, covering data cleaning, aggregation, and visualization across five progressively advanced scenarios, using the author's own code and generated graphs.

Project Overview

This project performs a structured, end-to-end data analysis of game reviews from the IGN dataset (**ign.csv**). The analysis is organized into five scenarios of increasing difficulty, moving from basic data handling to advanced multi-graph analysis, so as to provide comprehensive coverage of data preprocessing, aggregation, and visualization skills.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the ign.csv dataset.

Primary libraries used:

- **Pandas** — for data loading, cleaning, filtering, grouping, and aggregation.
- **NumPy** — for array conversion and yearly score-growth calculations (`np.diff`).
- **Matplotlib** — for all visualizations: line graphs, bar charts, pie charts, stacked bar charts, and histograms.

Project Coverage

Category	Skills Covered
Data Preprocessing	Data loading, missing-value handling (mean/mode), type conversion, column removal
Pandas Operations	Grouping, filtering, aggregation, feature engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, yearly score-growth calculation (<code>np.diff</code>)
Visualization	Line chart, bar chart, pie chart, histogram, stacked bar chart, saving plots

Scenario 1: Basic Data Loading & Cleaning

The dataset is loaded and inspected, then cleaned so it is ready for the analysis steps that follow: the redundant index column is dropped, missing values are handled, and column types are corrected.

Task	Description
1	Load the ign.csv dataset using Pandas.
2	Display the first 5 rows (head()), last 5 rows (tail()), and the shape of the dataset.
3	Remove the unnecessary column: "Unnamed: 0".
4	Check for missing values in score, genre, and platform.
5	Handle missing values: fill the numeric column score with the mean, and the categorical column genre with the mode.
6	Ensure correct data types: convert score → float and release_year, release_month, release_day → integer.

Code

```
df = pd.read_csv("ign.csv")

print(df.head())
print(df.tail())
print("Shape of dataset:", df.shape)

# Remove unnecessary index column
if "Unnamed: 0" in df.columns:
    df = df.drop(columns=["Unnamed: 0"])

# Check missing values
print(df[["score", "genre", "platform"]].isnull().sum())

# Handle missing values
df["score"] = df["score"].fillna(df["score"].mean())
df["genre"] = df["genre"].fillna(df["genre"].mode()[0])

# Ensure correct data types
df["score"] = df["score"].astype(float)
df["release_year"] = df["release_year"].astype(int)
df["release_month"] = df["release_month"].astype(int)
df["release_day"] = df["release_day"].astype(int)
```

Conclusion

- The dataset contains 18,625 rows and 11 columns, covering game titles, review scores, genres, platforms, and release dates.
- The redundant "Unnamed: 0" index column was removed to keep the feature set clean and consistent.
- Missing values existed in the score, genre, and platform columns and were imputed — the numeric score column with its mean, and the categorical genre/platform columns with their mode.
- Column data types were corrected (score → float; release_year/month/day → integer) so the data could be aggregated and plotted reliably in later steps.

Scenario 2: Line Graph — Score Trend Over Years

A line graph is constructed to show how the average review score has trended across release years, after grouping the data by `release_year` and computing the mean score per year.

Task	Description
1	Group data by <code>release_year</code> .
2	Calculate the average score per year using Pandas.
3	Convert the resulting average scores and years into NumPy arrays.
4	Plot a line graph with <code>release_year</code> on the X-axis and average score on the Y-axis.
5	Add the title: "Average Game Score Over Years" and proper axis labels.
6	Save the graph: <code>plt.savefig("avg_score_trend.png")</code> .

Code

```
yearly_avg_score = df.groupby("release_year")["score"].mean()

years = np.array(yearly_avg_score.index)
avg_scores = np.array(yearly_avg_score.values)

plt.figure(figsize=(10, 6))
plt.plot(years, avg_scores, marker="o", color="steelblue")
plt.title("Average Game Score Over Years")
plt.xlabel("Release Year")
plt.ylabel("Average Score")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("graphs/avg_score_trend.png")
```

Graph

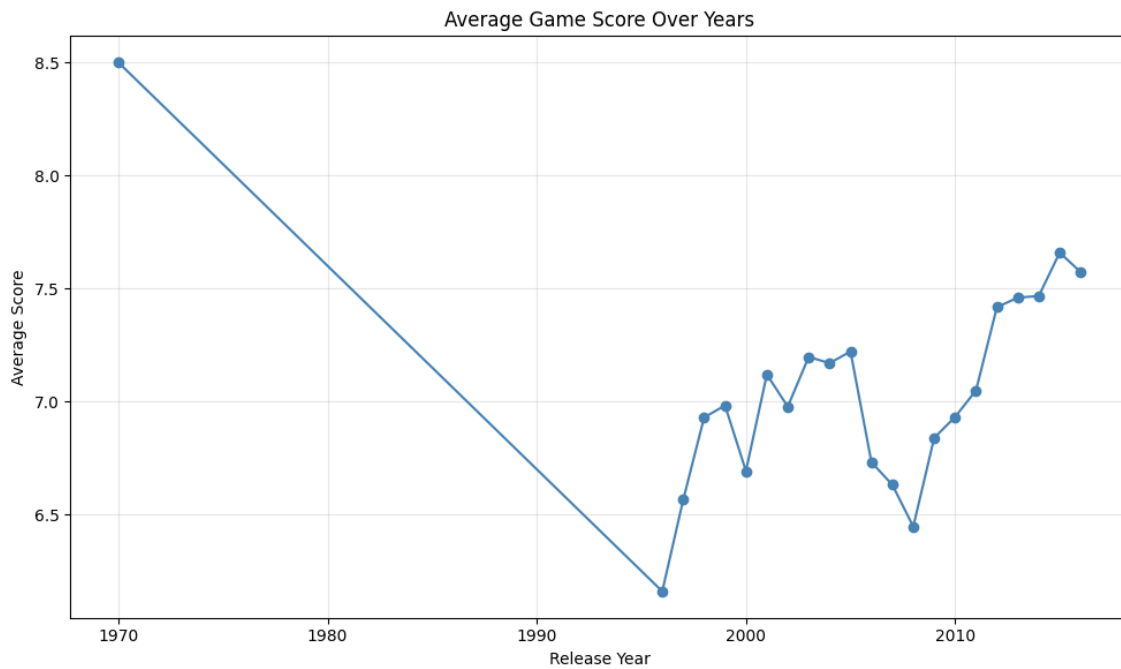


Fig 2.1 — Average Score Trend Over Years

Conclusion

- In 1970 the average score is a single high outlier around 8.5, before dropping sharply to its lowest point of roughly 6.2 by 1996.
- From the late 1990s onward, scores recover steadily, fluctuating moderately between about 6.5 and 7.2 through the 2000s.
- From 2010 onward, average scores climb consistently, reaching close to 7.5–7.7 by 2015–2016 — the highest sustained levels in the dataset.
- Overall, the trend shows a long recovery followed by steady improvement in review scores in recent years.

Scenario 3: Filtering + Bar Chart — Top Platforms

The dataset is filtered to high-rated games (score > 7) to identify which platforms carry the most well-reviewed titles, and the top 10 platforms are visualized as a bar chart.

Task	Description
1	Filter the dataset to include only games where score > 7.
2	Count the number of high-rated games per platform.
3	Select the top 10 platforms using Pandas.
4	Convert the platform counts into NumPy arrays.
5	Plot a bar chart with platform on the X-axis and count of games on the Y-axis.
6	Rotate x-axis labels for readability.
7	Save the graph: plt.savefig("top_platforms_bar.png").

Code

```
highRated = df[df["score"] > 7]
platformCounts = highRated["platform"].value_counts().head(10)

platforms = np.array(platformCounts.index)
counts = np.array(platformCounts.values)

plt.figure(figsize=(10, 6))
plt.bar(platforms, counts, color="seagreen")
plt.title("Top 10 Platforms by Count of High-Rated Games (score > 7)")
plt.xlabel("Platform")
plt.ylabel("Count of Games")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.savefig("graphs/top_platforms_bar.png")
```

Graph

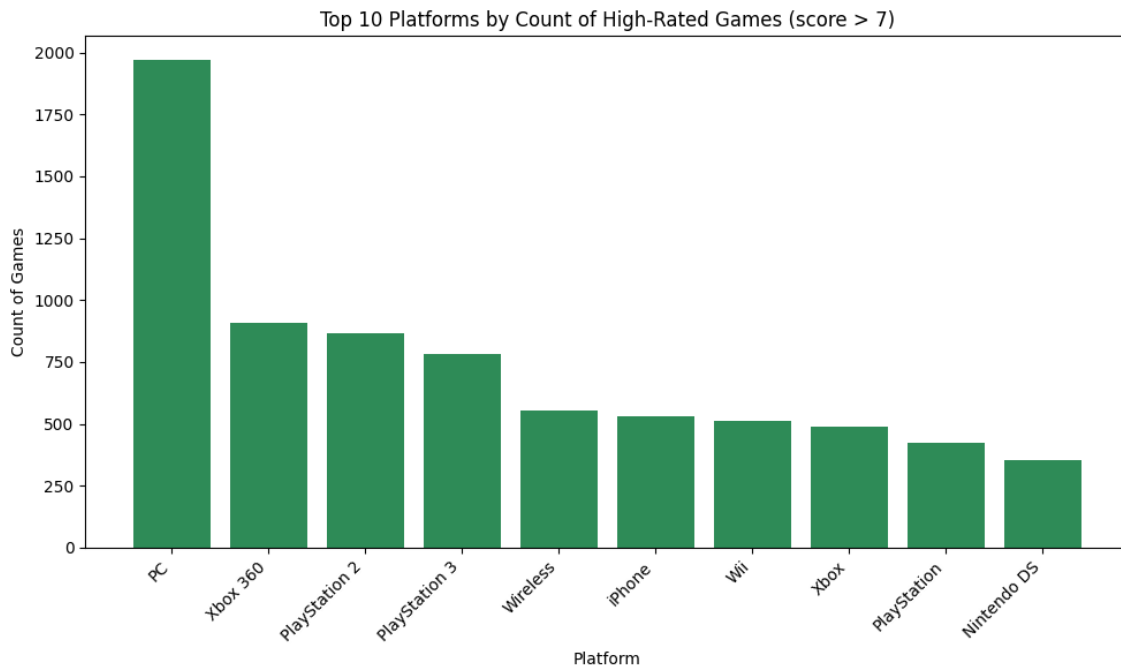


Fig 3.1 — Top 10 Platforms by High-Rated Game Count

Conclusion

- PC dominates the list with roughly 1,970 high-rated games — far ahead of every other platform.
- Xbox 360 ranks second (~905 games), followed by PlayStation 2 (~865) and PlayStation 3 (~780), showing strong performance among consoles.
- Mid-tier platforms — Wireless, iPhone, and Wii — each contribute a moderate share of high-rated titles.
- PlayStation and Nintendo DS have the lowest counts among the top 10, at the bottom of the ranking.
- Overall, modern and widely adopted platforms tend to have more high-rated games than older or niche ones.

Scenario 4: Aggregation + Pie Chart — Genre Distribution

Games are aggregated by genre to identify the top 5 most common genres, which are then visualized as a pie chart showing their relative share of the dataset.

Task	Description
1	Count the total number of games per genre.
2	Select the top 5 genres using Pandas.
3	Prepare labels (genres) and values (counts) for the pie chart.
4	Plot a pie chart using genre as labels and count as values.
5	Add percentage labels using autopct (e.g., '%.1f%%').
6	Save the graph: plt.savefig("genre_distribution.png").

Code

```
genre_counts = df["genre"].value_counts().head(5)

labels = genre_counts.index
values = genre_counts.values

plt.figure(figsize=(8, 8))
plt.pie(values, labels=labels, autopct="%1.1f%%", startangle=90,
        colors=plt.cm.Pastel1.colors)
plt.title("Top 5 Genre Distribution")
plt.tight_layout()
plt.savefig("graphs/genre_distribution.png")
```

Graph

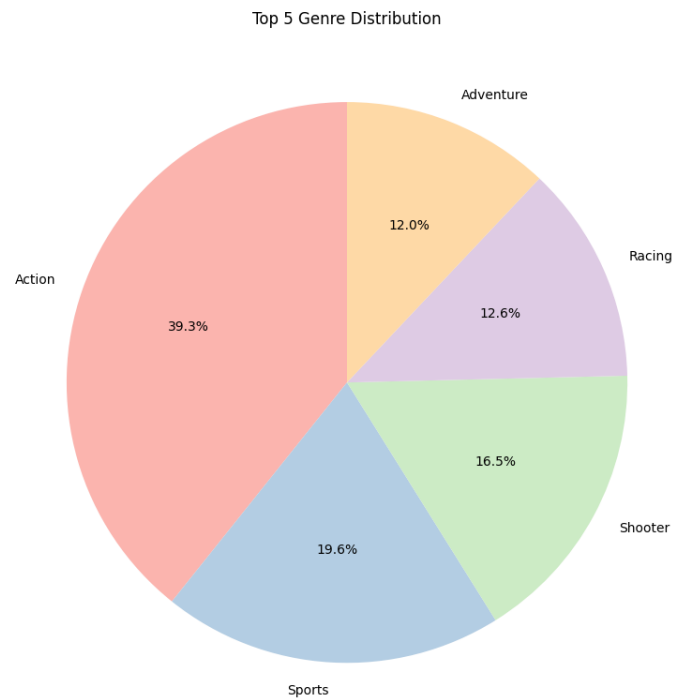


Fig 4.1 — Top 5 Genre Distribution

Conclusion

- Action is by far the most dominant genre, accounting for 39.3% of the top-5 total — nearly double the next closest genre.
- Sports is the second most common genre at 19.6%.
- Shooter games make up 16.5%, showing a strong and consistent presence.
- Racing (12.6%) and Adventure (12.0%) form a similarly sized smaller tier.
- Overall, the chart highlights a heavy skew toward Action titles, while the remaining top genres are more evenly distributed among themselves.

Scenario 5: Advanced Analysis + Multiple Graphs

This final scenario adds feature engineering — a `score_category` column (Excellent / Good / Average) and a numeric encoding of `editors_choice` — followed by NumPy-based yearly growth calculations and three visualizations: a line graph, a stacked bar chart, and a histogram.

Task	Description
1	Create a new column <code>score_category</code> : "Excellent" for <code>score > 9</code> , "Good" for $7 \leq \text{score} < 9$, and "Average" for <code>score < 7</code> .
2	Convert the <code>editors_choice</code> column: 'Y' → 1, 'N' → 0.
3	Use NumPy's <code>np.diff()</code> to calculate the yearly score growth based on the average yearly scores from Scenario 2.
4	Plot a Line Graph showing the trend of average score per <code>release_year</code> .
5	Plot a Stacked Bar Chart showing the count of <code>score_category</code> per <code>release_year</code> .
6	Plot a Histogram showing the distribution of <code>score</code> .
7	Save all three graphs to file.
8	Identify which years had the highest scores, whether high scores increased over time, and whether <code>editors_choice</code> correlates with high scores.

Code

```
def categorize_score(score):
    if score >= 9:
        return "Excellent"
    elif score >= 7:
        return "Good"
    else:
        return "Average"

df["score_category"] = df["score"].apply(categorize_score)
df["editors_choice"] = df["editors_choice"].map({"Y": 1, "N": 0})

# NumPy yearly growth
yearly_avg_score = df.groupby("release_year")["score"].mean()
yearly_scores_np = np.array(yearly_avg_score.values)
yearly_growth = np.diff(yearly_scores_np)

# Line graph
plt.plot(yearly_avg_score.index, yearly_avg_score.values, marker="o", color="darkorange")
plt.savefig("graphs/score_trend.png")

# Stacked bar chart
category_by_year = pd.crosstab(df["release_year"], df["score_category"])
category_by_year.plot(kind="bar", stacked=True, colormap="viridis")
plt.savefig("graphs/score_category_stacked.png")

# Histogram
plt.hist(df["score"], bins=20, color="mediumpurple", edgecolor="black")
plt.savefig("graphs/score_distribution.png")

# Insights
```

```
best_year = yearly_avg_score.idxmax()
corr = df["editors_choice"].corr(df["score"])
```

Graphs

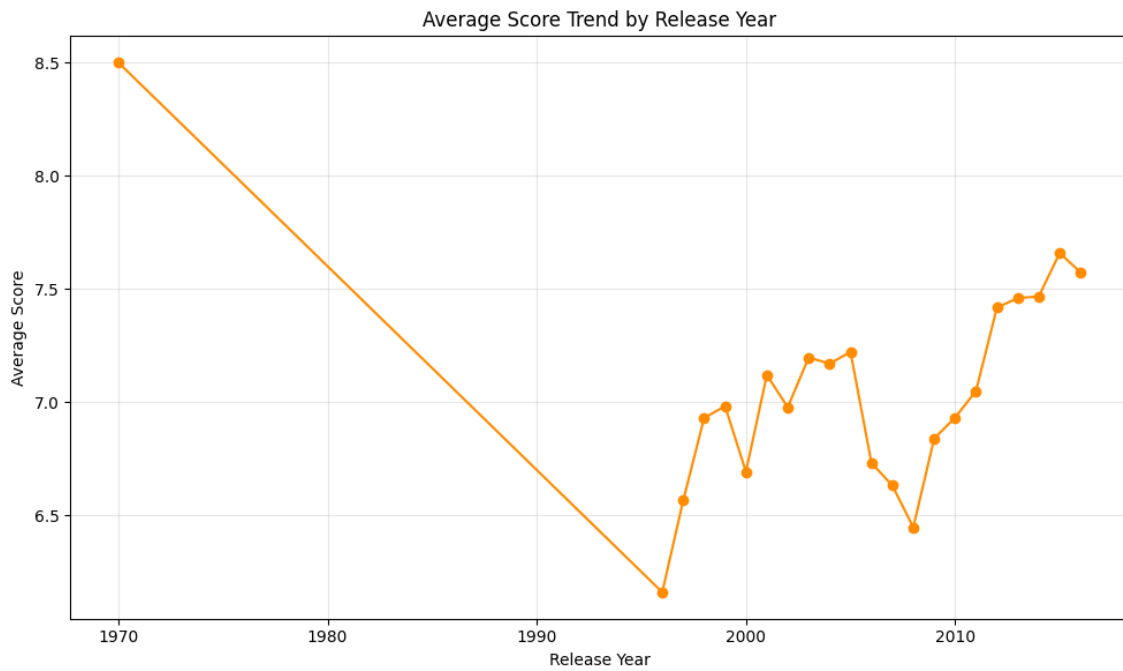


Fig 5.1 — Average Score Trend by Release Year

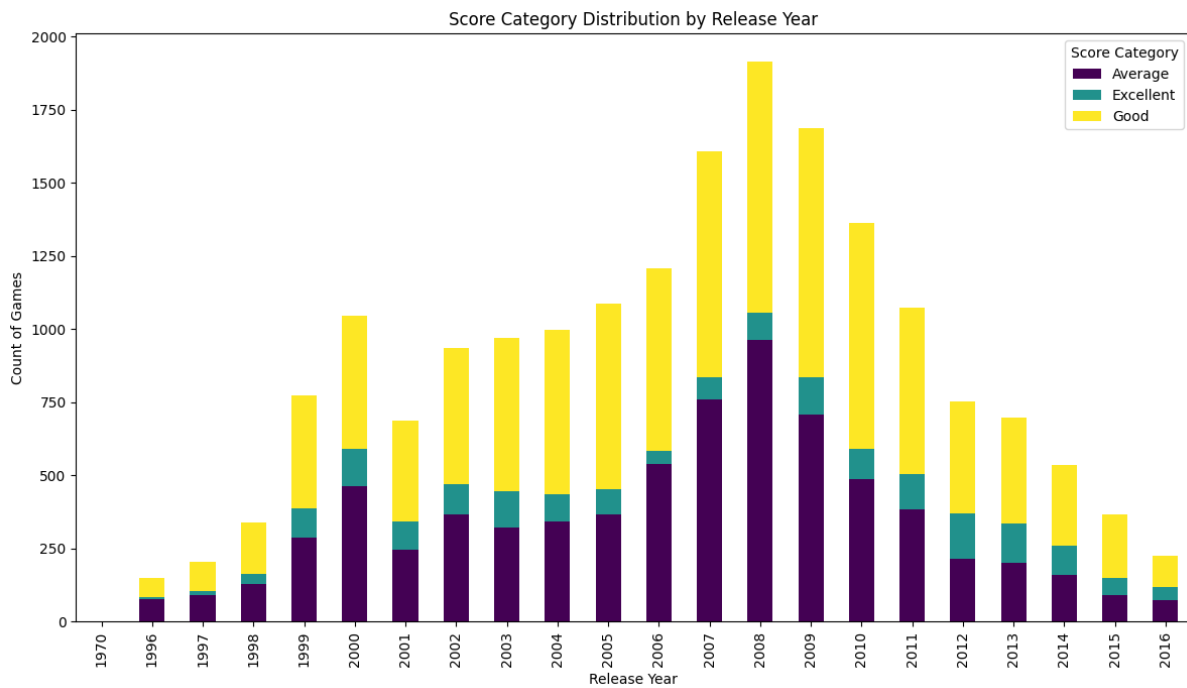


Fig 5.2 — Score Category Distribution by Release Year

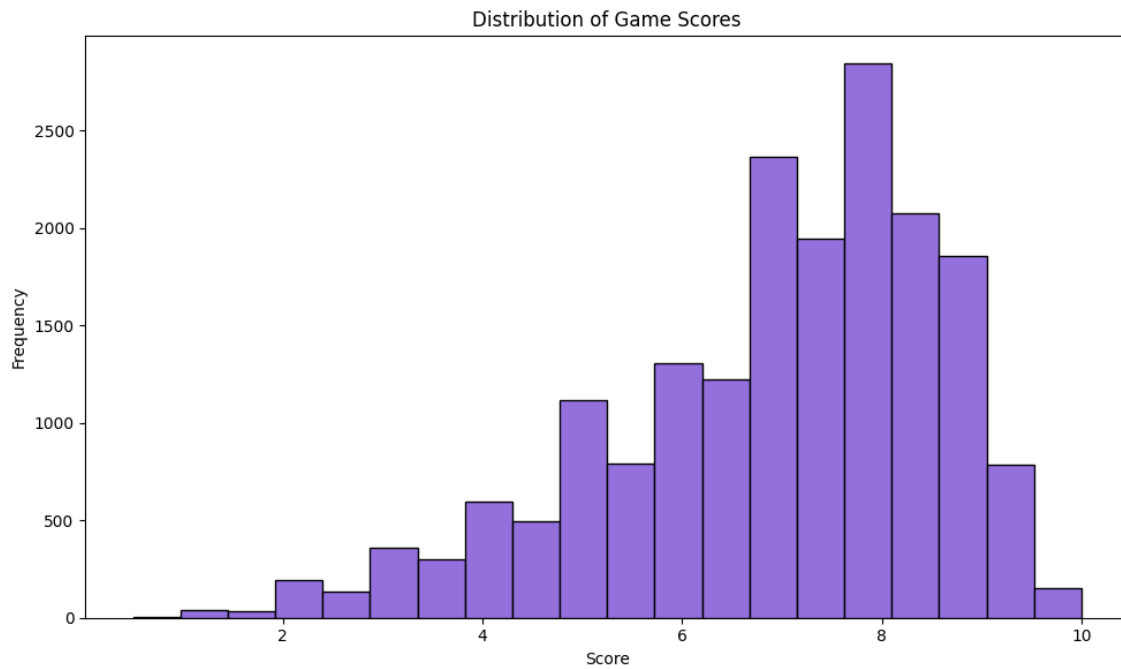


Fig 5.3 — Distribution of Game Scores (Histogram)

Conclusion

- The line chart confirms a sharp decline in average scores from 1970 to the mid-1990s, followed by gradual recovery and a steady rise after 2000, stabilizing around 7.5+ in recent years.
- The stacked bar chart shows the yearly count of Average, Good, and Excellent games, with total game production peaking between 2006 and 2009.
- The histogram of scores is slightly left-skewed toward higher values, peaking between 7 and 8, meaning most games are moderately-to-well rated; very low or very high scores are relatively rare.
- The editors_choice column shows a positive correlation with score, indicating that Editor's Choice titles tend to be rated notably higher than average.
- Together, the three charts indicate rising game production alongside a gradual improvement in overall game quality since the early 2000s.

Final Conclusion

Across the full analysis, a clear pattern emerges in both game quality and production volume over time. The average score trend shows an initial decline through the mid-1990s, followed by a steady recovery and consistent improvement in later years. The score histogram confirms that most games fall in the moderate-to-high range, peaking around 7–7.5, with extremely high or low scores being comparatively rare — suggesting a fairly consistent reviewing standard over time.

The score-category-by-year breakdown highlights a marked increase in the number of games released, especially between 2005 and 2009, spanning the Average, Good, and Excellent categories. The platform analysis shows that PC and the major consoles contribute the most high-rated games, while the genre analysis reveals a strong dominance of Action titles. Overall, these insights suggest that the gaming industry grew in both volume and quality over the period covered by this dataset, with review standards improving alongside more consistent output on the most popular platforms and genres.

What I Learned

This project reinforced practical data-analysis skills: encoding categorical values into numeric form for downstream use, engineering new columns that classify rows by condition (such as `score_category`), and using NumPy for calculations like yearly score growth. Building the visualizations also made clear how much more effectively charts communicate hidden patterns and trends compared to reading the raw data alone.